Vectors in n Dimensions

ANSWER KEY

Section 1: Vector Arithmetic and Linear Combinations

1 Given $\mathbf{u} = [2, -1, 4]$ and $\mathbf{v} = [-3, 5, 2]$ in $\mathbb{R}^3$, compute $3\mathbf{u} - 2\mathbf{v}$.

a. $[12, -13, 8]$ ✓

b. $[0, 7, 16]$

c. $[12, -7, 8]$

d. $[-12, 13, -8]$

$3\mathbf{u} = [6, -3, 12]$. $-2\mathbf{v} = [6, -10, -4]$. Sum $= [6+6, -3-10, 12-4] = [12, -13, 8]$.

2 Is the vector $\mathbf{w} = [7, 4]$ a linear combination of $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [3, -1]$?

a. **Yes, $\mathbf{w} = \frac{19}{7}\mathbf{v}_1 + \frac{10}{7}\mathbf{v}_2$** ✓

b. No, the vectors are linearly dependent

c. Yes, $\mathbf{w} = 2\mathbf{v}_1 + \mathbf{v}_2$

d. No, because the determinant is zero

Solve $c_1[1, 2] + c_2[3, -1] = [7, 4] \Rightarrow c_1 + 3c_2 = 7$ and $2c_1 - c_2 = 4 \Rightarrow c_2 = 2c_1 - 4$. Substitute: $c_1 + 6c_1 - 12 = 7 \Rightarrow 7c_1 = 19 \Rightarrow c_1 = 19/7, c_2 = 10/7$.

3 What is the definition of the span of a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ in $\mathbb{R}^n$?

a. **The set of all possible linear combinations $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k$ where $c_i \in \mathbb{R}$** ✓

b. The set of all vectors perpendicular to $\mathbf{v}_1$

c. The sum of the lengths of the vectors

d. The matrix formed by using the vectors as columns

By definition, $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is the subspace consisting of all linear combinations of those vectors.

Section 2: Linear Independence and Dependence

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4 A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is linearly independent if and only if:

a. $c_1\mathbf{v}_1 + \ldots + c_k\mathbf{v}_k = \mathbf{0}$ **implies** $c_1 = c_2 = \ldots = c_k = 0$ ✓

b. At least one vector is a multiple of another

c. The dot product of any two vectors is zero

d. The number of vectors k is greater than the dimension n

Linear independence requires that the only linear combination yielding the zero vector is the trivial solution where all coefficients are zero.

5 Can any set of 4 vectors in $\mathbb{R}^3$ be linearly independent?

a. **No, any set of more than n vectors in $\mathbb{R}^n$ is guaranteed to be linearly dependent** ✓

b. Yes, provided none of the vectors is the zero vector

c. Yes, if they all have unit length

d. Only if their cross products are non-zero

By the Fundamental Theorem of Linear Algebra, any collection of $k > n$ vectors in $\mathbb{R}^n$ must be linearly dependent because there are at most n pivot columns.

6 Are the vectors $\mathbf{v}_1 = [1, 2, 0]$, $\mathbf{v}_2 = [0, 1, 3]$, and $\mathbf{v}_3 = [2, 5, 3]$ linearly independent?

a. **No, because $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$** ✓

b. Yes, they span all of $\mathbb{R}^3$

c. No, because they contain a zero entry

d. Yes, because their determinant is 1

Check: $2\mathbf{v}_1 + \mathbf{v}_2 = 2[1, 2, 0] + [0, 1, 3] = [2, 4 + 1, 3] = [2, 5, 3] = \mathbf{v}_3$. Since one vector is a combination of the others, the set is dependent.

Section 3: Dot Product, Norm, and Orthogonality

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7 Compute the Euclidean norm (length) of the vector $\mathbf{v} = [2, -3, 6]$ in $\mathbb{R}^3$.

a. 7 ✓

b. $\sqrt{11}$

c. 49

d. $\sqrt{41}$

$\|\mathbf{v}\| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7.$

8 Find the unit vector in the direction of $\mathbf{u} = [-1, 2, 2]$.

a. $\left[-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$ ✓

b. $[-1, 2, 2]$

c. $\left[-\frac{1}{9}, \frac{2}{9}, \frac{2}{9}\right]$

d. $\left[-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right]$

$\|\mathbf{u}\| = \sqrt{(-1)^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = 3$. The unit vector is $\frac{\mathbf{u}}{\|\mathbf{u}\|} = [-1/3, 2/3, 2/3]$.

9 What does the Cauchy-Schwarz inequality state for any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$?

a. $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$ ✓

b. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

c. $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|$

d. $|\mathbf{u} \cdot \mathbf{v}| \geq \|\mathbf{u}\|\|\mathbf{v}\|$

The Cauchy-Schwarz inequality states $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$, with equality holding if and only if the vectors are collinear.

Section 4: Orthogonal Projections

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10 What is the formula for the orthogonal projection of vector $\mathbf{y}$ onto a non-zero vector $\mathbf{u}$?

a. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}$ ✓

b. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{\mathbf{y}\cdot\mathbf{u}}{\|\mathbf{y}\|}\mathbf{u}$

c. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = (\mathbf{y}\cdot\mathbf{u})\mathbf{u}$

d. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \mathbf{y} - \mathbf{u}$

The projection of $\mathbf{y}$ onto the line spanned by $\mathbf{u}$ is $\frac{\mathbf{y}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\mathbf{u} = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}$.

11 Compute the projection of $\mathbf{y} = [7, 6]$ onto $\mathbf{u} = [4, 2]$.

a. $[8, 4]$ ✓

b. $[4, 2]$

c. $[7, 6]$

d. $[2, 1]$

$\mathbf{y}\cdot\mathbf{u} = 7(4) + 6(2) = 28 + 12 = 40$. $\mathbf{u}\cdot\mathbf{u} = 4^2 + 2^2 = 20$. $\text{proj}_{\mathbf{u}}(\mathbf{y}) = \frac{40}{20}[4, 2] = 2[4, 2] = [8, 4]$.

12 If $\hat{\mathbf{y}} = \text{proj}_{\mathbf{u}}(\mathbf{y})$, what is the geometric relationship between $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ and $\mathbf{u}$?

a. **$\mathbf{z}$ is orthogonal to $\mathbf{u}$** ✓

b. $\mathbf{z}$ is parallel to $\mathbf{u}$

c. $\mathbf{z}$ has unit length

d. $\mathbf{z} = \mathbf{0}$ always

By the Orthogonal Decomposition Theorem, any vector can be uniquely decomposed into a component parallel to $\mathbf{u}$ and an orthogonal component $\mathbf{z} \perp \mathbf{u}$.

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